

Euclid: Precision astrometry with Euclid VIS: a transferable focal-plane correction from Gaia-CRF3.

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ABSTRACT

Context: High-precision astrometry with the *Euclid* Visible Instrument (VIS) requires an accurate characterisation of detector-level systematic distortions at the milliarcsecond level.

Aims: We investigate the astrometric performance of *Euclid* Q1 VIS single calibrated-frame catalogues by comparing source positions with the inertial optical reference frame defined by Gaia-CRF3. Our aim is to test whether the structured residuals in VIS astrometry contain a stable, detector-fixed component that can be modelled globally and transferred to independent observations.

Methods: We construct a sample of 9425 clean Gaia-CRF3 quasi-stellar objects in the *Euclid* Deep Fields, covering the full VIS focal plane of 36 CCDs and 144 quadrants. The resulting displacement field is spatially coherent, with amplitudes increasing from the focal-plane centre towards the edges, indicating that the dominant limitation of the nominal single-frame astrometry is a stable geometric distortion rather than random centroiding noise. We model this component in detector coordinates with an independent second-order polynomial vector field per quadrant.

Results: In an independent ROS3609 validation field, the correction reduces the median absolute deviation from 4.46 mas to 3.03 mas in right ascension and from 6.69 mas to 2.28 mas in declination. The performance is competitive with both the *Euclid* DR1 validation products and an independent field-by-field recalibration, demonstrating that a single transferable global model can approach the precision of local solutions. After the geometric correction, a colour-dependent residual becomes visible in the broader stellar validation sample, traced with the Gaia $G_{BP} - G_{RP}$ index and structured across the detector. A first-order colour correction recentres the residual distributions to below 0.1 mas, motivating a future joint calibration in colour and focal-plane position. As an illustrative demonstration, the correction is applied to non-Q1 VIS observations. These contain the streak of asteroid 66146 (1998 TU3), and our method shifts measured endpoints by 1 mas – 2 mas; the remaining comparison with JPL Horizons is dominated by timing and trajectory uncertainties.

Conclusions: These results establish *Euclid* VIS as a dense, Gaia-referenced astrometric layer for faint-source applications, provided geometric and chromatic systematics are modelled jointly.

Note : This draft is intended as a starting point for scientific and authorship review before submission.

Key words. astrometry – methods: data analysis – reference systems – instrumentation: detectors – surveys – minor planets, asteroids: individual: 66146 (1998 TU3)

1. Introduction

Astrometry underpins both the construction of celestial reference frames and a wide range of dynamical measurements, from Galactic structure to Solar System orbits. The current optical realisation of the International Celestial Reference System is provided by Gaia, whose third celestial reference frame, Gaia-CRF3, is based on extragalactic sources and provides an effectively inertial optical reference system (Gaia Collaboration: Klioner et al. 2022). The *Euclid* mission, although primarily designed for cosmology, offers high-resolution, wide-field imaging that can transfer this reference frame to fainter and denser source populations than are available directly from Gaia (Euclid Collaboration: Mellier et al. 2024).

The VIS instrument combines a large focal plane with a broad optical passband and a small pixel scale. This makes it valuable for astrometric applications, but also makes the calibration sensitive to residual geometric distortions, detector-dependent effects, and chromatic centroid shifts. The scientific question addressed here is whether the residuals between *Euclid* VIS positions and Gaia-CRF3 contain a stable detector-fixed component that can be learned globally and applied to independent observations. If such a component exists, it can provide a

practical astrometric correction layer for VIS single-frame catalogues and improve their usefulness as local reference catalogues for fainter objects.

This paper, based on a MSc thesis project, is organized as follows. Section 2 describes the *Euclid* and Gaia data used. Section 3 presents the detector-space distortion model. Section 4 summarises the astrometric validation and the emergence of chromatic residuals. Section 5 gives a moving-object demonstration using asteroid 66146 (1998 TU3). Section 6 discusses the limitations and prospects.

2. Data and reference frame

The analysis uses *Euclid* Q1 VIS calibrated-frame catalogues in the *Euclid* Deep Fields. The astrometric reference sample is drawn from Gaia-CRF3, whose quasi-stellar objects provide a stable optical frame with negligible parallax and proper motion for the purposes of this calibration. Starting from the cross-matched VIS–Gaia sample, catalogue flags, magnitude cuts, and separation cuts are used to remove problematic detections, blends, and uncertain matches. The final calibration sample contains 9425 clean Gaia-CRF3 matches.

The residuals are first defined on the sky as

$$\Delta\alpha^* = (\alpha_{\text{Gaia}} - \alpha_{\text{VIS}}) \cos \delta, \quad \Delta\delta = \delta_{\text{Gaia}} - \delta_{\text{VIS}}, \quad (1)$$

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where the Gaia positions are used as the external astrometric reference. Fig 1 shows the residual cloud is centred near zero but shows milliarcsecond-level scatter and structured tails. For modelling the instrumental distortion component, the residuals are transformed into detector coordinates. This is convenient for applying the static detector-fixed distortion as it projects differently onto right ascension and declination for observations taken at different orientations.

3. Detector-space distortion model

The focal-plane displacement field is sampled over all 36 VIS CCDs and 144 quadrants. The residual vectors show a coherent pattern over the full mosaic: amplitudes grow from the centre towards the outer quadrants, and the vector orientations change smoothly across the focal plane. Typical median displacements are approximately 0.02–0.03 pixels in central quadrants and reach 0.20–0.26 pixels in the extreme corners. This morphology points to a low-order, stable geometric distortion rather than source-by-source random noise.

We model the displacement field independently in each quadrant. For quadrant i , normalised detector coordinates (x', y') are formed by subtracting the mean source position in that quadrant and dividing by the corresponding coordinate standard deviations. The correction is then represented by a second-degree vector polynomial,

$$f_x^{(i)}(x', y') = a_0^{(i)} + a_1^{(i)}x' + a_2^{(i)}y' + a_3^{(i)}x'^2 + a_4^{(i)}x'y' + a_5^{(i)}y'^2, \quad (2)$$

$$f_y^{(i)}(x', y') = b_0^{(i)} + b_1^{(i)}x' + b_2^{(i)}y' + b_3^{(i)}x'^2 + b_4^{(i)}x'y' + b_5^{(i)}y'^2. \quad (3)$$

The corrected detector position is obtained by subtracting the fitted displacement field from the measured VIS position:

$$\Delta x \equiv x_{\text{CRF3}} - x_{\text{VIS}}, \quad \Delta y \equiv y_{\text{CRF3}} - y_{\text{VIS}}, \quad (4)$$

A second-order model contains 12 free parameters per quadrant, while the calibration sample provides two constraints per source. This keeps the fit constrained while allowing enough flexibility to capture smooth local distortions. No continuity constraint is imposed across quadrant boundaries; the smoothness of the recovered solution is therefore an empirical result rather than an enforced prior.

4. Validation and chromatic residuals

The model is validated on the ROS3609 field by comparing corrected VIS positions against Gaia DR3 stellar positions propagated to the epoch of the *Euclid* observation. This test uses a broader and denser validation sample than the Gaia-CRF3 quasar calibration set. It therefore probes both the transferability of the geometric correction and the residual systematics that appear for realistic stellar populations.

The nominal VIS calibrated-frame catalogue already reaches the few-milliarcsecond regime, but its residuals contain the detector-fixed structure described above. After applying the global correction, the median absolute deviations improve from 4.46 mas to 3.03 mas in $\Delta\alpha^*$ and from 6.69 mas to 2.28 mas in $\Delta\delta$. The corrected performance is competitive with *Euclid* Data Release 1 validation products (private communication). And with an independent field-by-field recalibration, while requiring only a transferable global model (Libralato et al. (2024)).

Once the dominant geometric term is removed, a second systematic becomes apparent: the residuals depend on source colour (see Fig ??). Using the Gaia $G_{\text{BP}} - G_{\text{RP}}$ colour index, the

post-correction stellar residuals show a clear trend with colour, whereas the quasar calibration sample has a narrower colour distribution and is therefore less sensitive to this effect. A first-order colour correction applied independently in right ascension and declination recentres the residual distributions and reduces the mean offset to below 0.1 mas.

Detector-coordinate stacking confirms that the chromatic term is not merely a sky-coordinate artefact: it has a spatial pattern across the focal plane. This indicates that the next stage of the calibration should combine source colour and detector position in a single model, ideally using *Euclid*-native colour information from NISP rather than relying only on Gaia colours. This is beyond the scope of this paper.

5. Application to asteroid 66146 (1998 TU3)

As a practical demonstration, the correction is applied to the VIS streak of asteroid 66146 (1998 TU3) in ROS4564. The moving-object case differs from the static-source calibration because the measured quantity is not a stellar centroid but a streak endpoint, and the comparison with an external ephemeris depends on timing, trajectory interpolation, and observer geometry.

The B26 correction shifts the inferred streak endpoint positions by approximately 1 mas – 2 mas, consistent with the expected amplitude of the local focal-plane model (Fig 4). However, the remaining discrepancy with JPL Horizons is larger than the correction itself and is dominated by timing and trajectory uncertainties rather than by the focal-plane distortion term. The asteroid test, like the validation on ROS3609, is again consistent with the idea that the correction method is applicable outside the Q1 dataset. Unfortunately, the milliarcsecond focal-plane improvements matter only once the full external error budget on observation absolute timing and asteroid orbit prediction is controlled at a comparable level.

6. Discussion

The results show that a stable geometric component is present in *Euclid* Q1 VIS single-frame astrometry and that it can be corrected with a transferable detector-space model. The improvement in ROS3609 demonstrates that the model is not merely fitting noise in the calibration sample. This is important for applications in which *Euclid* can act as an intermediate astrometric layer between Gaia and deeper, higher-resolution facilities. Gaia anchors the bright end of the reference frame, while corrected *Euclid* catalogues can provide a denser set of fainter reference objects.

The present model has clear limitations. It corrects the detector-fixed geometric component but does not re-optimize field-level shifts, rotations, or scale terms in the world-coordinate solution. It is calibrated on quasars, which provide an excellent inertial reference but span a relatively narrow colour range. Consequently, chromatic residuals become visible when the model is validated on a broader stellar population. The colour correction presented here is deliberately empirical and first-order; it is sufficient to demonstrate the effect but not yet a full operational chromatic calibration.

A natural next step is a joint model in detector position and colour. *Euclid* NISP photometry is the most attractive route because VIS and NISP observe the same fields through the same telescope at the same epoch. NISP-based colours would be more closely tied to the *Euclid* instrumental response than Gaia

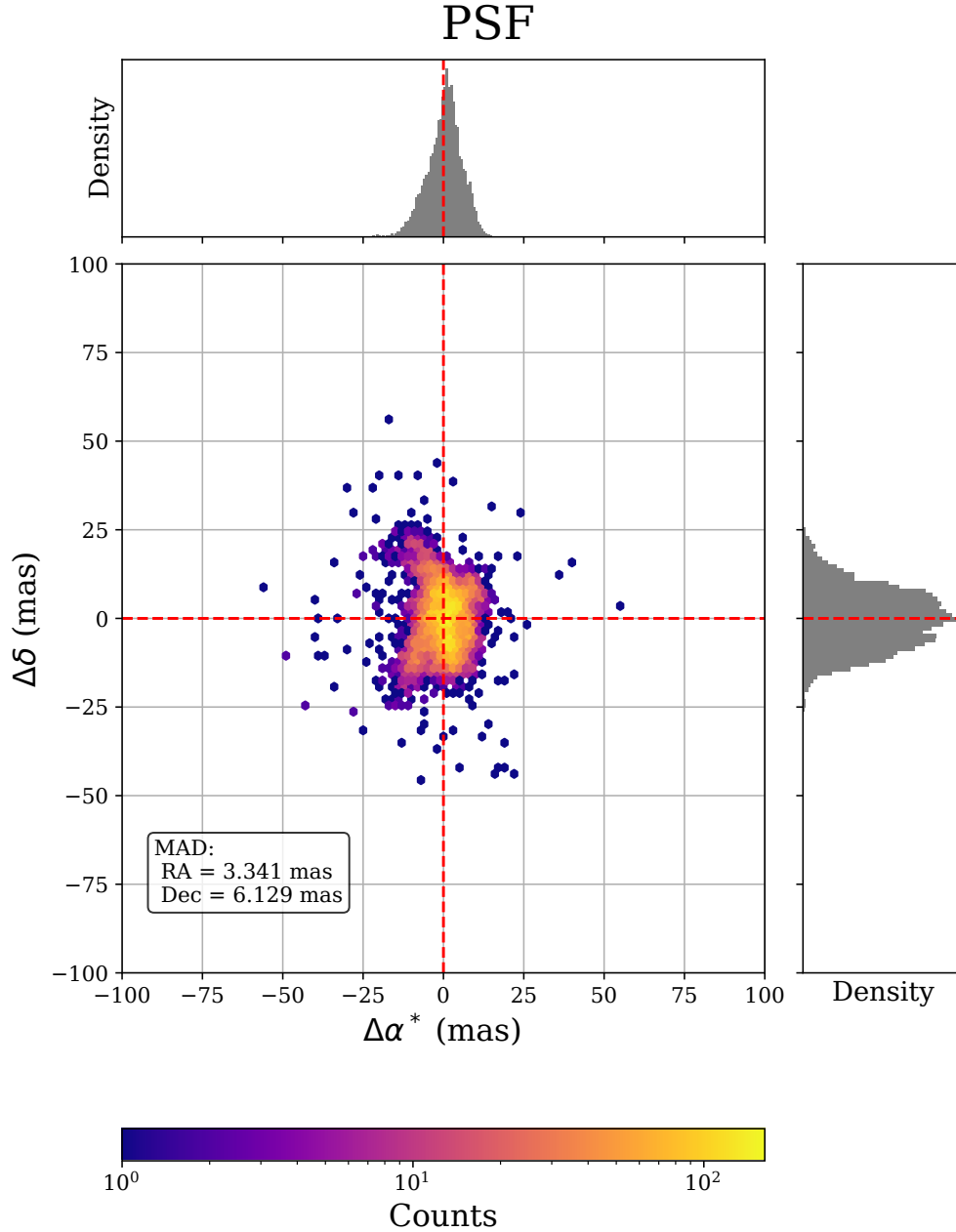


Fig. 1. Astrometric residuals between *Euclid* VIS PSF-fitting positions and Gaia-CRF3 positions for the matched Q1 sample of 9425 Gaia-CRF3 objects before the detector-level correction. The residual cloud is centred near zero but shows milliarcsecond-level scatter and structured tails.

$G_{BP} - G_{RP}$ and would extend the calibration to fainter sources. Such a model should be trained on large samples of compact, isolated, unsaturated sources and should use robust aggregate statistics because the chromatic displacement is small compared with the source-by-source scatter.

7. Conclusions

We have distilled the analysis into the following main conclusions:

1. The residual field between *Euclid* Q1 VIS single-frame positions and Gaia-CRF3 is spatially coherent across the VIS focal plane, with amplitudes increasing towards the detector edges.
2. A per-quadrant second-order polynomial vector model in detector coordinates captures a stable, transferable geometric distortion component.
3. In the independent ROS3609 validation field, the correction improves the median absolute deviations from 4.46 mas to 3.03 mas in $\Delta\alpha^*$ and from 6.69 mas to 2.28 mas in $\Delta\delta$.
4. After the geometric correction, a colour-dependent residual becomes visible in the broader stellar validation sample. A first-order colour correction recentres the mean residuals to below 0.1 mas, but a full calibration should combine colour and focal-plane position.
5. Application to asteroid 66146 (1998 TU3) shifts measured endpoints by 1 mas – 2 mas; the remaining discrepancy with the predicted position by JPL Horizons is set by timing and trajectory uncertainties rather than by the geometric correction.

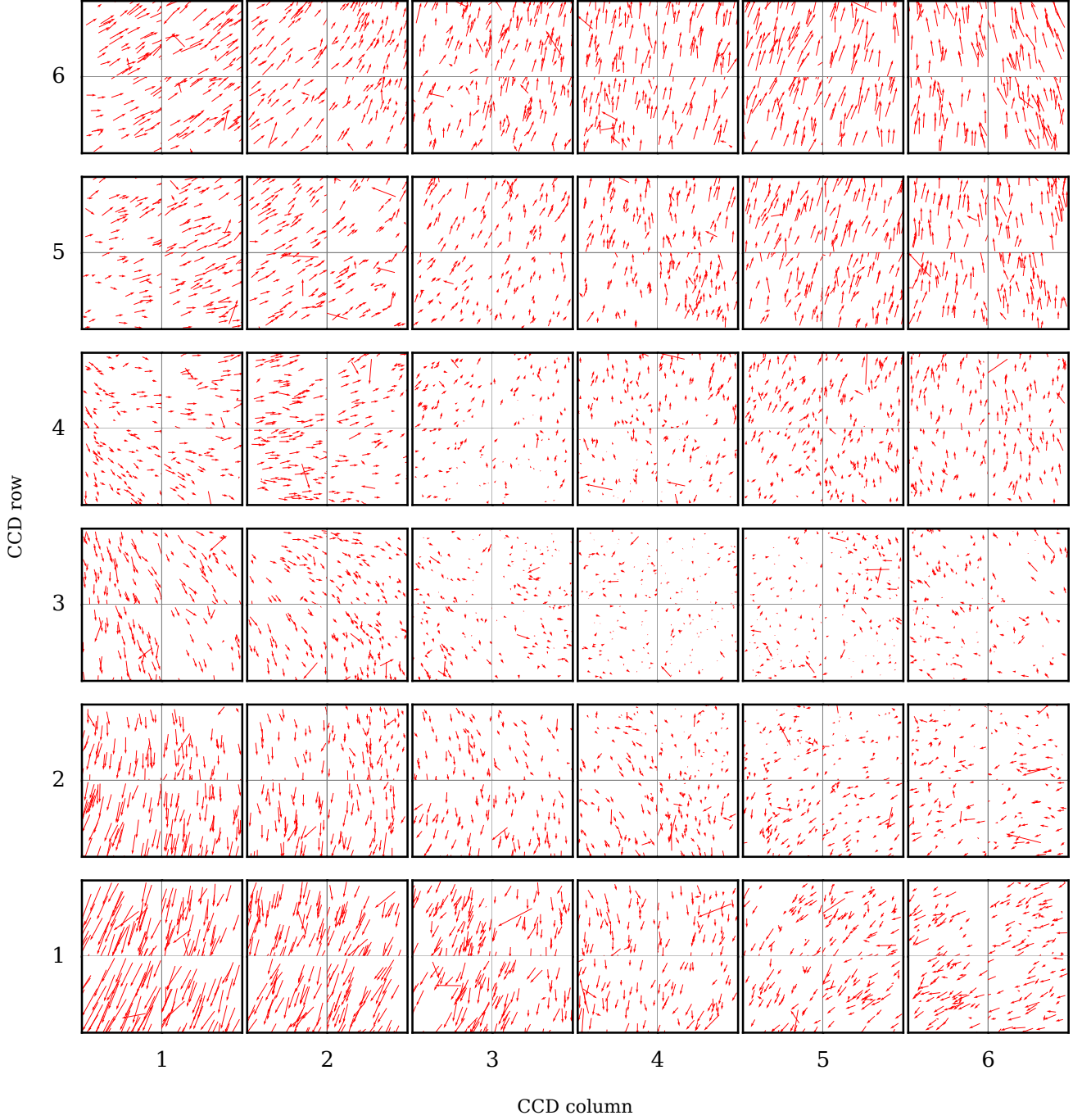


Fig. 2. Pixel-space displacement vectors between Gaia CRF3 and Euclid VIS measurements. The figure shows the CRF3–VIS matched sources across all Q1 fields, combining measurements from several hundred VIS exposures on the VIS focal plane. For each CRF3–VIS matched source, a vector is drawn with origin at $(x_{\text{CRF3}}, y_{\text{CRF3}})$, and orientation and length given by $(\Delta x, \Delta y)$, defined in equation 4. Vectors are magnified by a factor of 3000.

These results support the use of corrected *Euclid* VIS catalogues as a Gaia-referenced, dense astrometric layer at fainter magnitudes, while identifying chromaticity and field-level alignment as the next limiting systematics.

References

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ROS3609

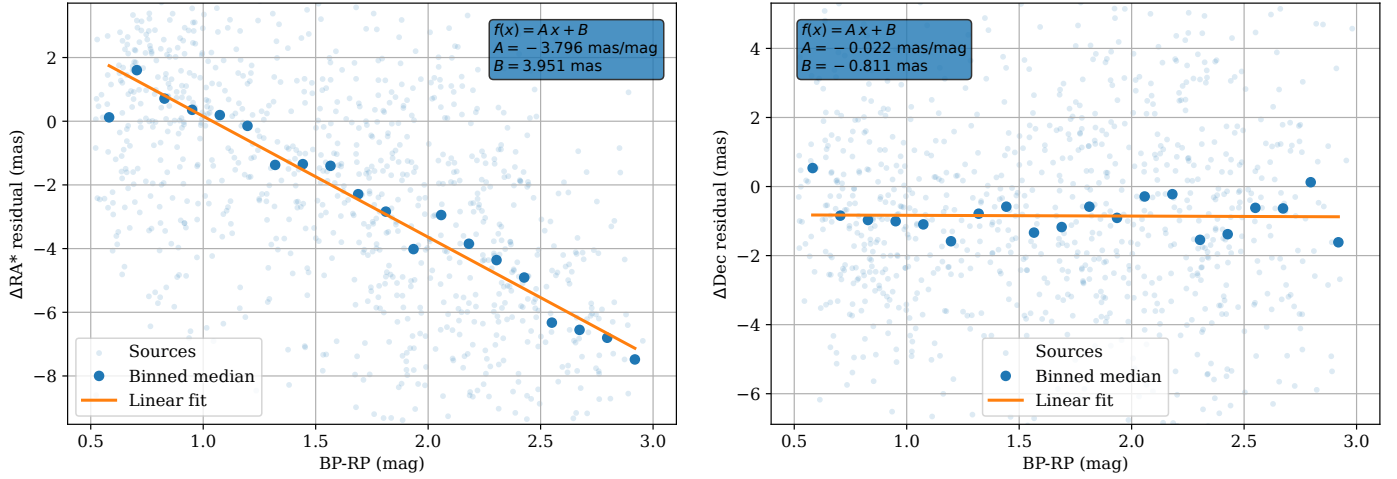


Fig. 3. Chromatic correction for ROS3609. The upper panel shows the astrometric residuals in $\Delta\alpha^*$ and $\Delta\delta$ as a function of Gaia colour BP – RP, together with the best-fitting linear models derived independently for the two components. The lower panels compare the two-dimensional residual distributions before (left) and after (right) applying the colour correction. The correction reduces the colour-dependent trend and recentres the residual distribution around the origin.

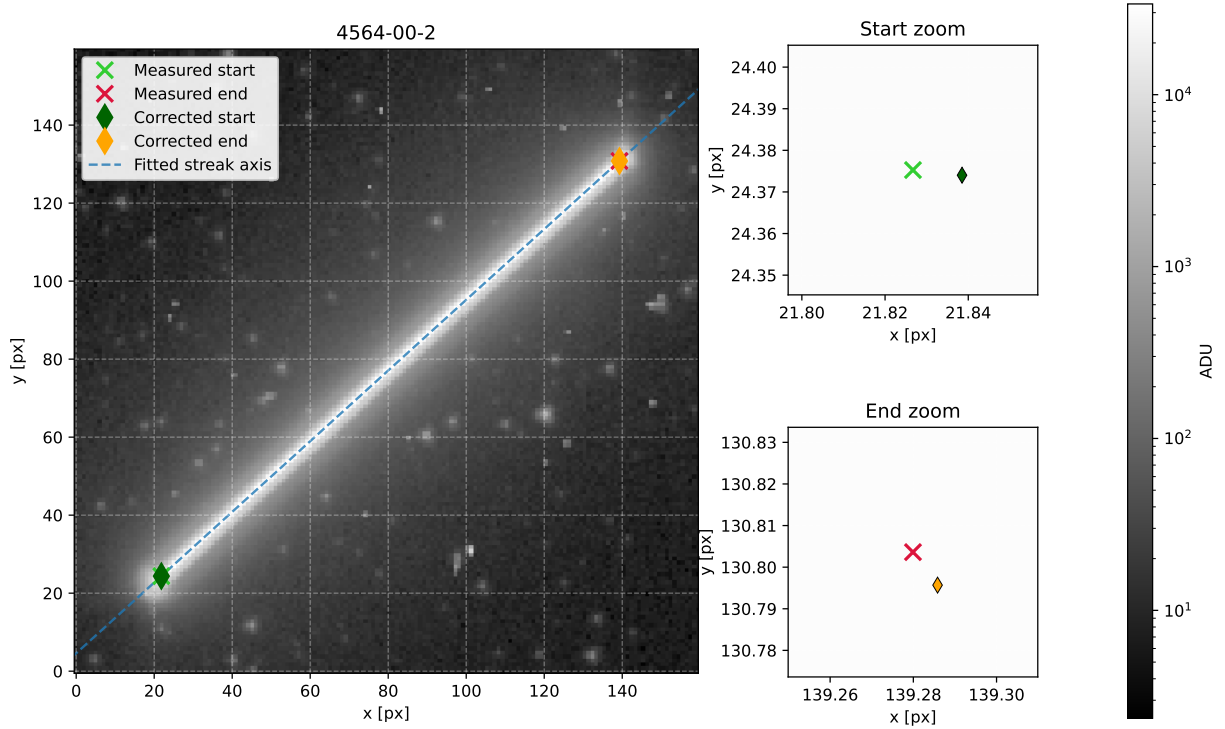


Fig. 4. Endpoint measurements for exposure 4564-00-2. The main panel shows the streak cutout with the fitted streak axis and the measured start and end positions before and after applying the B26 correction. The two smaller panels provide zoomed views of the start and end regions, highlighting the local displacement introduced by the correction.

Appendix A: FAIR access to analysis code, model coefficients and full quadrant statistics and other details

205 For Findability, Accessibility, Interoperability and Reusability of the results we refer to the original MSc thesis report. For example, it contains the full per-quadrant distortion coefficients and residual statistics, access to code repository used to derive the results presented in this paper plus additional detailed analysis. The MSc thesis is available at XX.